

THE COPPERBELT UNIVERSITY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

TOPIC: Multi-Source Energy-Harvesting Autonomous Sensor Node.

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INTRODUCTION

Wireless sensor networks are widely used in environmental monitoring, smart agriculture, infrastructure surveillance, and disaster management. However, most deployed sensor nodes rely on batteries, which limit system lifetime and require periodic replacement. This project proposes a battery-less autonomous sensor node powered entirely by ambient energy harvesting to address these challenges.

1.1 Background of the Study

Battery dependency introduces high maintenance costs, logistical challenges in remote areas, and environmental concerns. Energy harvesting offers a sustainable alternative by extracting energy from ambient sources such as sunlight gradients, mechanical radio frequency vibrations, and electromagnetic fields. However, single-source harvesting systems suffer from unreliability due to environmental variability. A multi-source harvesting approach improves robustness and energy availability.

1.2 Problem Statement

Conventional sensor nodes depend on batteries and often a single energy source, leading to frequent maintenance, unreliable operation, and limited deployment feasibility in remote or hazardous environments.

1.3 Objectives

Main Objective: To design and implement a multi-source ambient energy harvesting autonomous sensor node.

Specific Objectives:

1. To analyze suitable ambient energy harvesting techniques.
2. To design an efficient power management and storage subsystem.
3. To implement adaptive duty-cycling algorithms.

1.4 Scope and Applicability

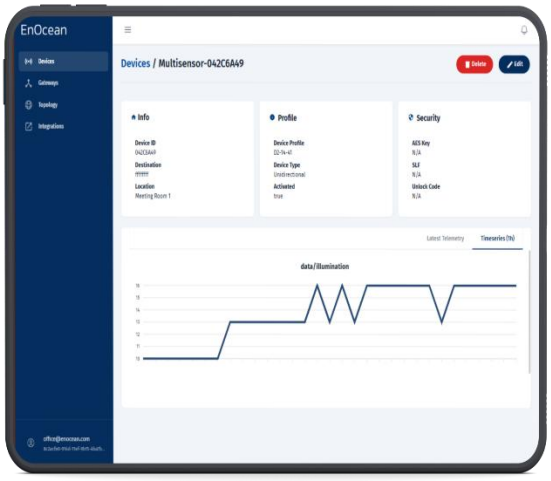
The project focuses on the design and prototyping of autonomous sensor node for low power sensor nodes, that can be deployed on different fields such as: agricultural sector, mining, military, weather focus stations for environmental monitoring or infrastructure surveillance.

LITERATURE REVIEW

2 Previous Systems (or Similar Applications)

This section reviews existing systems and applications that are functionally related to the proposed **multi-source energy-harvesting, battery-less autonomous sensor node**. The reviewed systems provide energy harvesting, sensing, monitoring, or management capabilities similar to the proposed solution.

2.1 EnOcean Device Management Platform





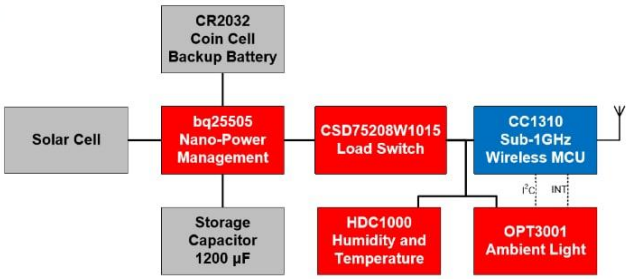
- **System name:** EnOcean Device Management Platform
- **URL:** https://www.enocean.com
- **Platform:** Web-based (Windows, Linux, macOS via browser)
- **Trial license:** Demo kits (vendor-controlled)

Description:

This platform was developed to configure and monitor EnOcean self-powered sensors used in building automation. It targets building managers and system integrators seeking maintenance-free sensor deployments. While reliable, it employs conservative duty cycles and single-source harvesting, limiting adaptability compared to the proposed system.

2.2 Texas Instruments Energy Harvesting Evaluation GUI (BQ25570)







- **System name:** TI Energy Harvesting Evaluation GUI
- **URL:** https://www.ti.com
- **Platform:** Windows
- **Trial license:** Free evaluation software

Description:

Developed to support evaluation of TI energy-harvesting PMICs, this application visualizes voltage levels and MPPT behavior. It targets embedded system designers and researchers. The software focuses on hardware characterization and lacks autonomous sensing and system-level intelligence.

2.3 Power cast RF Energy Harvesting Monitoring System



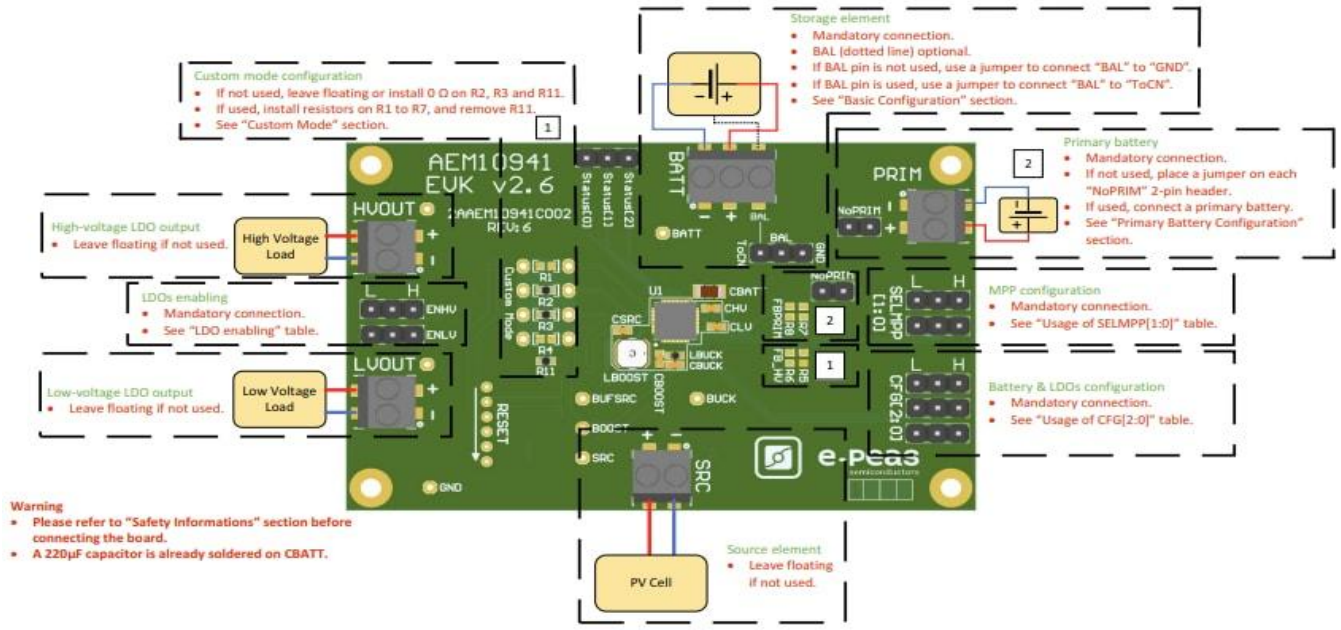


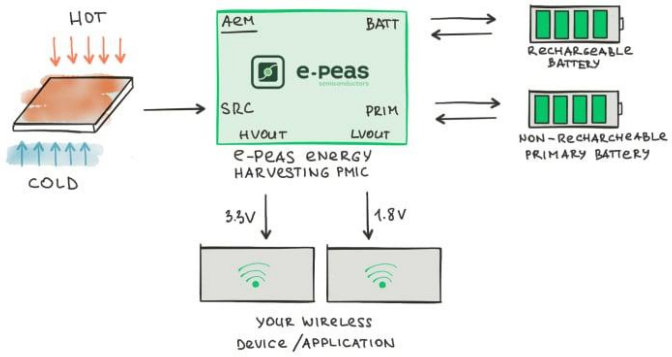
- **System name:** Power cast Energy Monitoring Software
- **URL:** https://www.powercastco.com
- **Platform:** Web-based / Windows
- **Trial license:** Hardware evaluation kits (time-limited)

Description:

This system monitors RF-powered IoT devices in wireless power environments. It targets engineers deploying RF-powered sensors. Dependence on dedicated RF transmitters and lack of hybrid harvesting limit its flexibility

2.4 E-peas Energy Harvesting Demonstration Software (AEM10941)





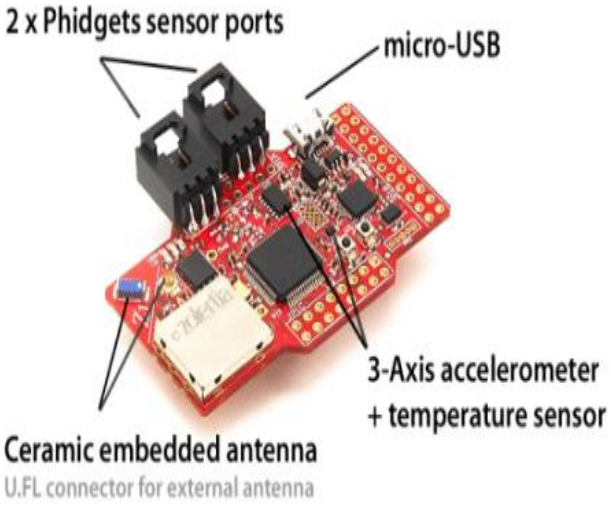
- **System name:** e-peas Energy Harvesting Demo Software
- **URL:** https://www.e-peas.com
- **Platform:** Windows
- **Trial license:** Evaluation kits

Description:

Designed to demonstrate photovoltaic harvesting with MPPT and super capacitor storage, this software targets hardware engineers. It is hardware-centric and does not implement predictive or adaptive task scheduling.

2.5 Zolertia IoT Energy-Harvesting Node Dashboard





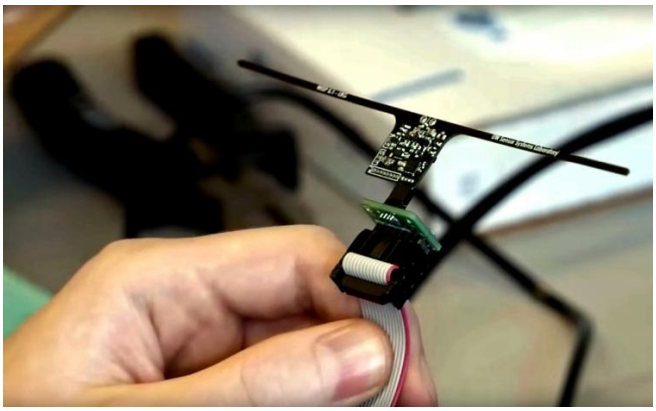
- **System name:** Zolertia IoT Management Dashboard
- **URL:** https://zolertia.io
- **Platform:** Web-based
- **Trial license:** Open-source / research use

Description:

This dashboard supports monitoring of low-power IoT nodes in research deployments. Target users include researchers and universities. Most nodes remain battery-assisted, limiting full autonomy.

2.6 Wireless Identification and Sensing Platform (WISP) Tools





- **System name:** WISP Monitoring and Debug Tools
- **URL:** https://wisp.wikispaces.com
- **Platform:** PC-based research tools
- **Trial license:** Open research platform

Description:

Developed to demonstrate battery-free RF-powered sensing, WISP targets academic researchers. Operation is constrained by reader proximity and intermittent power.

2.7 Nordic Semiconductor BLE Energy Profiling Tools

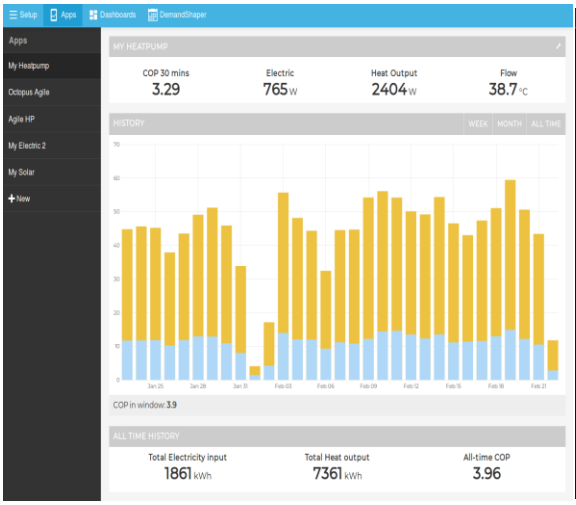


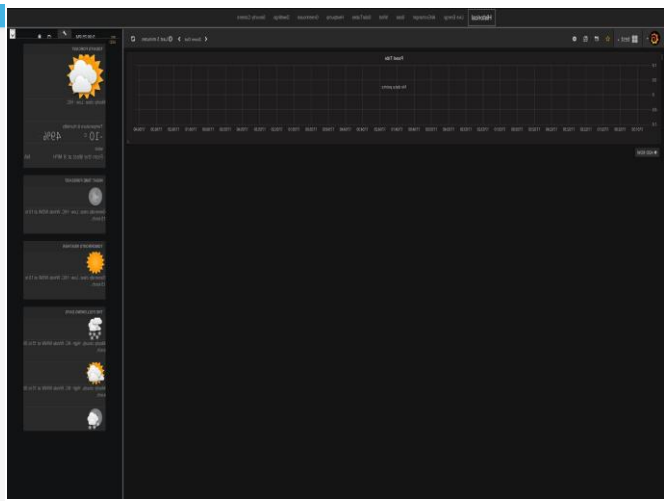
- **System name:** Nordic Power Profiler
- **URL:** https://www.nordicsemi.com
- **Platform:** Windows
- **Trial license:** Free with hardware

Description:

This tool analyzes power consumption of BLE devices and targets embedded developers. It does not provide harvesting or autonomous operation but highlights communication energy dominance.

2.8 Open Energy Monitor IoT Monitoring Platform



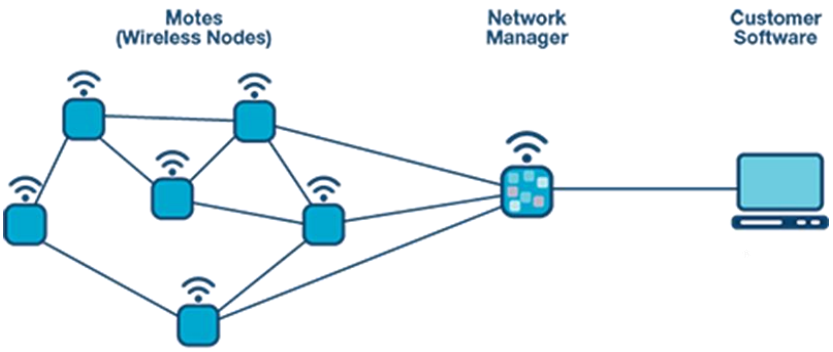


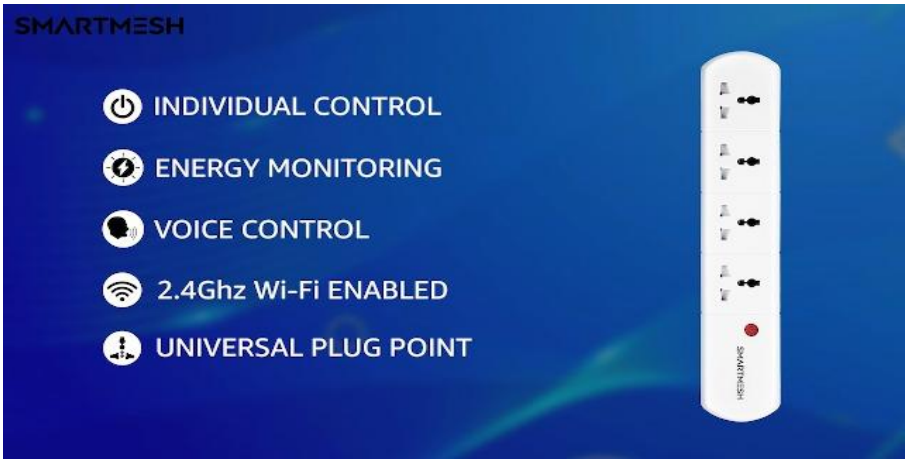
- **System name:** Open Energy Monitor
- **URL:** https://openenergymonitor.org
- **Platform:** Web-based
- **Trial license:** Open-source

Description:

Originally developed for energy monitoring, this platform supports sensor data visualization. It targets hobbyists and researchers. Battery-less harvesting is not native, limiting suitability for autonomous sensing.

2.9 Smart Mesh IP Power Monitoring Tools



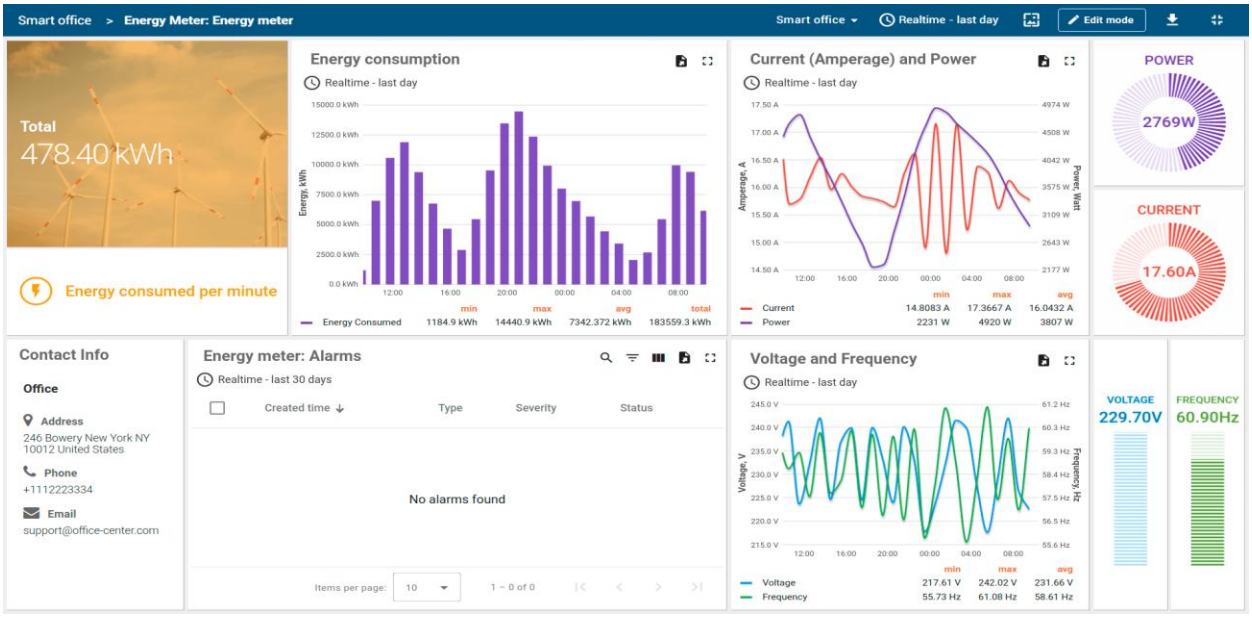


- **System name:** Smart Mesh Network Manager
- **URL:** https://www.analog.com
- **Platform:** Web-based / PC
- **Trial license:** Evaluation-based

Description:

This system manages ultra-low-power wireless sensor networks. It targets industrial IoT users. Nodes typically rely on batteries rather than ambient harvesting

2.10 Things Board IoT Monitoring Platform



- **System name:** Things Board
- **URL:** https://thingsboard.io
- **Platform:** Web-based
- **Trial license:** Community Edition (free)

Description:

Things Board is a generic IoT monitoring platform used to visualize sensor data. It targets IoT developers and enterprises. Energy harvesting awareness and adaptive duty cycling are not natively supported.

3 Ambient Energy Harvesting Techniques

Research by Paradiso and Sterner demonstrated the feasibility of powering low-power electronics using ambient energy. Solar, piezoelectric, and thermoelectric harvesting techniques have been widely explored, though each suffers from environmental dependency namely.

3.1 Solar Energy Harvesting

Solar energy harvesting is one of the most extensively researched and practically deployed ambient energy harvesting techniques due to its comparatively high energy density and technological maturity. It employs photovoltaic (PV) cells to convert incident solar radiation directly into electrical energy through the photovoltaic effect **(Paradiso & Starner, 2005)**. Because solar radiation provides higher power levels than most other ambient sources, PV harvesting is frequently adopted as the primary energy source for outdoor and semi-outdoor wireless sensor systems **(Roundy et al., 2004)**.

Experimental and analytical studies indicate that PV-based harvesters can produce output power ranging from a few microwatts under indoor or low-light conditions to several milli-watts under direct sunlight, depending on illumination intensity, panel surface area, material efficiency, and temperature **(Roundy et al., 2004)**. However, the electrical characteristics of PV cells are highly nonlinear. Variations in irradiance and temperature significantly affect output voltage and current, making direct coupling to electronic loads inefficient and potentially unstable.

To mitigate these effects, Maximum Power Point Tracking (MPPT) techniques are widely employed to dynamically adjust the operating point of the PV cell and maximize harvested energy under changing environmental conditions **(Esram & Chapman, 2007)**. Despite these advancements, solar energy harvesting remains inherently intermittent, as it depends on daylight cycles and weather conditions, limiting its reliability as a standalone energy source **(Paradiso & Starner, 2005)**.

3.2 Magnetic Field Energy Harvesting

Magnetic field energy harvesting exploits alternating magnetic fields generated by electrical infrastructure such as power transmission lines, transformers, motors, and industrial machinery **(Beeby & White, 2010)**. This technique typically utilizes inductive coils to convert time-varying magnetic flux into electrical energy based on Faraday's law of electromagnetic induction **(Roundy et al., 2004)**.

Magnetic energy harvesting is particularly advantageous in indoor and industrial environments where strong magnetic fields are continuously present. Unlike solar harvesting, it is independent of lighting conditions and can provide energy as long as the source equipment remains operational **(Beeby & White, 2010)**. This makes it attractive for environments where light availability is limited or inconsistent.

However, the harvested power levels are generally low and strongly dependent on coil geometry, distance from the magnetic source, operating frequency, and field strength **(Roundy et al., 2004)**. As a result, magnetic harvesting alone is insufficient to sustain continuous sensing and wireless communication. Research indicates that it is most effective as a supplementary energy source when combined with efficient rectification, voltage boosting, and intelligent energy management circuits **(Beeby & White, 2010)**.

3.3 Radio Frequency (RF) Energy Harvesting

Radio Frequency (RF) energy harvesting captures energy from ambient electromagnetic waves emitted by radio transmitters, television stations, cellular base stations, and Wi-Fi access points **(Paradiso & Starner, 2005)**. A typical RF harvesting system consists of an antenna, impedance matching network, rectifier, and energy storage element, commonly referred to as ***antenna***. **(Sample et al., 2008)**.

One advantage of RF harvesting is its availability in both indoor and outdoor environments, particularly in urban areas where RF signals are continuously present. However, the ambient RF

power density is extremely low, often in the nano-watt to microwatt range, making RF harvesting unsuitable as a primary energy source for most sensor applications **(Roundy et al., 2004)**.

Despite these limitations, RF harvesting plays an important role in battery-less and ultra-low-power systems. It is particularly effective for standby operation, periodic wake-up events, and trickle charging of energy storage components when combined with high-efficiency rectification and impedance matching **(Sample et al., 2008)**. Consequently, RF harvesting is most effective when integrated into multi-source energy harvesting architectures.

3.4 Energy Storage Technologies for Energy Harvesting

Energy storage is a fundamental component of energy harvesting systems due to the intermittent and unpredictable nature of ambient energy sources **(Priya & Inman, 2009)**. The primary function of storage is to buffer harvested energy and provide a stable and continuous power supply to electronic loads during periods of insufficient energy availability.

Rechargeable batteries, such as lithium-ion and lithium-polymer types, offer high energy density and support long-duration operation. However, they suffer from limited charge–discharge cycles, aging effects, leakage, and environmental concerns, which reduce their suitability for long-term, maintenance-free deployments **(Beeby & White, 2010)**.

Super capacitors, also known as ultra-capacitors, provide high power density, rapid charging capability, and extremely long cycle life, often exceeding millions of cycles without significant degradation **(Priya & Inman, 2009)**. These characteristics make them particularly suitable for battery-less energy harvesting systems. Consequently, super capacitors and thin-film storage devices are increasingly preferred, with power management integrated circuits (PMICs) playing a critical role in voltage regulation, energy protection, and system stability **(Beeby & White, 2010)**.

RESEARCH METHODOLOGY

3.1 Iterative Engineering Process Model

The research employs a **design-based and experimental methodology**, focusing on the development of a functional prototype and performance evaluation. The study follows an engineering design process consisting of system analysis, design, implementation, and testing.

Initially, a comprehensive literature review is conducted to understand existing energy harvesting technologies, power management strategies, and intelligent duty-cycling mechanisms. Based on the identified research gaps, system requirements are defined, followed by the design of the overall architecture and subsystem models. Hardware design is carried out using electronic design principles.

3.2 Methodological Stages

3.2.1 Stage One: Literature Review and Requirement Analysis

extensive review of existing academic and industrial energy harvesting systems was conducted to identify design gaps. Functional and non-functional requirements were derived from this review.

Functional Requirements

The proposed system shall:

1. Harvest ambient energy from solar, magnetic, and radio frequency sources.
2. Convert and regulate harvested energy using a power management integrated circuit **(PMIC)**.
3. Store harvested energy in a super capacitor-based storage unit.
4. Monitor energy levels and estimate available energy in real time.
5. Execute intelligent energy-aware control decisions based on energy availability.
6. Adapt the duty cycle of sensing and communication tasks dynamically.
7. Transmit sensor data wirelessly using a low-power radio module.
8. Enter low-power or sleep modes when energy is insufficient.

Non-Functional Requirements

The proposed system shall:

1. In cooperate hardware and software design that aid in the minimizing power consumption.
2. Operate autonomously without the use of a battery.
3. Ensure system reliability and prevent brown-out conditions.
4. Support long-term operation with minimal maintenance.
5. Maintain compact physical size and low system complexity.
6. Provide scalability for additional energy sources or sensors.

3.2.2 Stage Tow: System Architecture Design and Modeling

In the second stage, the overall system architecture of the proposed multi-source energy-harvesting autonomous sensor node is being designed. This stage is involving the definition of interactions among the energy harvesting units (solar, magnetic, and radio frequency), the power management subsystem, energy storage elements, the microcontroller unit, sensors, and the wireless communication module. System-level modeling is being performed to analyze energy flow, power conversion efficiency, and operational feasibility under varying ambient energy conditions. The architectural design is ensuring modularity to allow independent optimization of individual subsystems.

3.2.3 Stage Three: Hardware Design and Component Selection

During this stage, detailed hardware design and component selection are being carried out. Electronic components are being selected based on ultra-low-power characteristics, efficiency, availability, and suitability for energy harvesting applications. Circuit schematics are being developed to integrate the energy harvesting sources, PMIC, super capacitor-based energy storage, microcontroller, sensors, and communication modules. Following schematic verification, the hardware design is being implemented through printed circuit board (PCB) design, ensuring compact layout, reduced power losses, and reliable electrical connectivity.

3.2.4 Stage Four: Firmware Development and Energy-Aware Control Algorithms

At this stage, firmware for the autonomous sensor node is being developed. Embedded software is being designed to monitor energy availability, manage power states, and execute adaptive duty-cycling algorithms. The firmware is dynamically adjusting sensing, processing, and communication operations based on real-time energy levels stored in the super capacitor. Energy-aware scheduling techniques are being implemented to minimize power consumption, prevent brown-out conditions, and maintain stable system operation under fluctuating ambient energy supply.

3.2.5 Stage Five: System Integration, Testing, and Evaluation

In the final stage, all hardware and firmware components are being integrated into a complete functional prototype. The system is being tested under varying environmental and energy availability conditions to evaluate overall performance. Key performance metrics, including energy harvesting efficiency, system uptime, operational reliability, and autonomous functionality, are being measured and analyzed. The experimental results are being compared against the defined system requirements to validate the effectiveness of the proposed multi-source energy-harvesting autonomous sensor node.

3.3 Tools and Technologies to Be Used

In this study, a range of hardware and software tools and technologies are being utilized to support the design, implementation, and evaluation of the proposed multi-source energy-harvesting autonomous sensor node. Low-power microcontrollers are being employed to execute sensing, control, and communication tasks, while dedicated energy harvesting power management integrated circuits (PMICs) are being used to efficiently regulate and store energy harvested from ambient solar, magnetic, and radio frequency sources. Super capacitor-based energy storage technologies are being implemented to buffer harvested energy and ensure stable system operation under intermittent energy conditions.

Electronic design automation (EDA) tools will be used for circuit schematic development and printed circuit board (PCB) layout design to ensure compactness, reliability, and minimal power losses. Embedded firmware is being developed using C/C++ programming environments to implement energy-aware control logic and adaptive duty-cycling algorithms. For system testing and evaluation, measurement instruments such as multi-meters, oscilloscopes, and power analyzers are being used to monitor voltage levels, current consumption, and overall system performance. Data analysis and documentation tools are being employed to record experimental results and support performance evaluation and reporting throughout the study.

SIGNIFICANCE OF STUDY

This study is significant as it contributes to the development of **sustainable and autonomous wireless sensor systems** by reducing reliance on conventional batteries. By integrating solar, magnetic, and radio frequency (RF) energy harvesting with intelligent energy-aware control, the proposed system addresses key challenges associated with limited energy availability and maintenance in wireless sensor networks.

The study demonstrates how multi-ambient energy harvesting combined with efficient power management and predictive duty-cycling can improve system uptime, reliability, and energy efficiency. This is particularly important for applications in remote, industrial, and hard-to-access environments where battery replacement is costly or impractical.

Academically, the project enhances understanding of energy harvesting system design, power management integrated circuits (PMICs), and low-power embedded systems, providing practical insights for students and researchers. Technologically, the findings support the advancement of **battery-less Internet of Things devices**, contributing to environmentally friendly and long-term sensing solutions.

EXPECTED CONTRIBUTIONS & IMPLEMENTATION

This study is expected to contribute both academically and practically to the fields of Computer Science, Computer Engineering, and Information Systems. The key expected contributions include:

Educational and Research Value

The project provides a reference implementation for students and researchers interested in low power embedded systems, energy harvesting, and sustainable (IoT) technologies.

Design of a Battery-Less Sensor System

The study presents a practical design of a battery-less wireless sensor node powered by ambient solar, magnetic, and RF energy, demonstrating an alternative to conventional battery-dependent systems.

Integration of Multi-Ambient Energy Harvesting

The research contributes a system architecture that combines multiple ambient energy sources with efficient power management, improving energy availability and system reliability.

Application of Intelligent Energy Management

The implementation of energy-aware control and predictive duty-cycling strategies contributes to improved energy efficiency and stable system operation under fluctuating energy conditions.

Implementation

The implementation of the study follows a systematic engineering approach. Initially, an extensive literature review was conducted to understand existing energy harvesting techniques, power management strategies, and intelligent control methods. Based on this review, system requirements and specifications are defined. The system will then be designed using modular hardware architecture, integrating ambient energy harvesting sources, a power management integrated circuit (PMIC), super capacitor-based energy storage, a microcontroller, and a wireless communication module. Hardware implementation is carried out through circuit design and printed

circuit board (PCB) development. Firmware will be developed to implement intelligent energy-aware control and predictive duty-cycling algorithms. Finally, results will be analyzed, documented, and presented as part of the study outcomes.

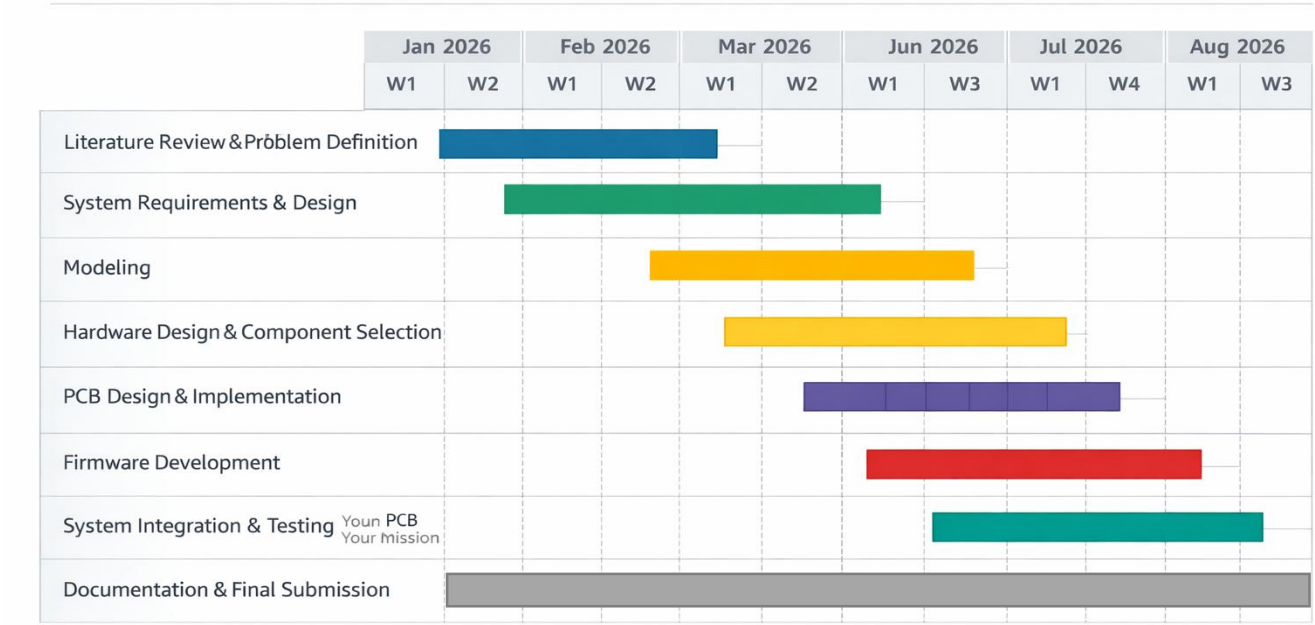
ETHICAL ISSUES IN (C.S, C.E, & I.S) RESEARCH

In accordance with this study, ethical considerations were carefully addressed throughout the design and implementation of the battery-less multi-ambient energy harvesting sensor node. The system was developed to operate autonomously without collecting personal or sensitive user data; therefore, data privacy and confidentiality risks are minimal. However, all system generated data were handled responsibly and used strictly for academic and evaluation purposes. Academic integrity was maintained by ensuring that all referenced theories, algorithms, and hardware designs were properly cited, avoiding plagiarism and intellectual property violations. Experimental results and performance evaluations were reported honestly and accurately without manipulation or exaggeration.

From an engineering perspective, the system is to be designed with **safety and reliability** in mind. Voltage regulation, energy threshold monitoring, and controlled duty cycling are implemented to prevent hardware damage, unstable operation, or unintended system behavior. Additionally, the battery-less design promotes **environmental responsibility** by reducing electronic waste and eliminating chemical battery disposal concerns. The study also considered the **social and technological impact** of the proposed system by promoting sustainable, low-power computing solutions suitable for long-term deployment. Overall, the research adhered to ethical standards in computer science and engineering by ensuring responsible system design, transparency, and societal benefit.

PROJECT TIMELINE

The project is being conducted from **January 2026 to August 2026** following a structured schedule. The study is starting with a literature review and problem definition in January to identify research gaps. System requirements and overall design are being developed between February and March. Modeling of system behavior and energy flow is being carried out from March to April. Hardware design and component selection are being undertaken between April and May, followed by PCB design and implementation in June. Firmware development is being performed in July. Finally, system integration, testing, documentation, and final submission are being completed in August 2026.



BUDGET

Item	Description	Estimated Cost (ZMW)
Photovoltaic (Solar) Cell	Small PV panel for energy harvesting	550
Magnetic Coil	Inductive coil for magnetic field harvesting	200
RF Antenna & Rectifier Components	RF energy harvesting circuitry	250
PMIC	Energy harvesting power management IC	750
Super capacitor	Energy storage element	300
Microcontroller	Low-power MCU (e.g., ESP32)	1000
Radio Module	Wireless communication module	300
Sensors	Environmental sensing components	650
PCB Fabrication	Two-layer PCB design and printing	500
Documentation	Printing, binding of SRS Documents	300
Miscellaneous	Wires, resistors, capacitors, connectors	500
Total Estimated Cost		**5,300 ZMW**

CONCLUSION

This study presented the design and implementation of a **battery-less intelligent energy-aware wireless sensor node** powered by ambient energy sources, namely solar, magnetic, and radio frequency (RF) energy. The primary objective of the project was to develop an autonomous sensor system capable of operating without conventional batteries by efficiently harvesting, managing, and utilizing ambient energy.

Through the integration of multi-ambient energy harvesting techniques, a power management integrated circuit (PMIC), super capacitor-based energy storage, and intelligent energy-aware control algorithms

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